

Anteater Poker Game (Version 1.0)

By All In

Members: Sanaa Bebal, Ava Chinn, Alonso De La Cruz, Estrella Distancia Zamora, Lauren Howe, Samuel Kim, and David Lee

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Glossary of Anteater Poker Terms

Game

- **Anteater Poker**—a version of poker based on Texas Hold'em, with the addition of the anteater cards and their rules (see “anteater card”)
- **Anteater card**—two exist in a deck; not a part of any standard suit; only a ranked hand when a player holds two anteaters in the hole cards, in which case it ranks between a royal and straight flush and four of a kind
- **Board**—a term used to refer to the visible community cards
- **Blinds**—the bids made by the player one left (the small blind) and two left (the big blind) of the dealer; small blind is 1 point and big blind is 2 points
- **Community cards**—the cards visible to all players that can be used by all players to create a five-card hand
- **Dealer**—the player in charge of giving cards to players; dealer deals clockwise
- **Five-card hand**—a group of five cards comprised of some combination of the two hole cards and the five community cards
- **Hand**—refers to the complete sequence of moves made within the repeatable segment of the game, starting with players receiving their cards and ending with someone winning the hand
- **Hole cards**—the two cards held by the individual player; only the player knows what their cards are and only the player can use them to create a five-card hand
- **Kicker**—refers to the cards that are not part of the ranked part of a hand that are used to break ties; used for ties involving one pair, two pair, three of a kind, and four of a kind
- **Pot**—holds the main points bet in a hand
- **Round**—consists of dealing cards and multiple betting intervals where the players match the highest bet, raise, or fold. Rounds include pre-flop, flop, turn, and river betting.
- **Side pot**—not the main pot, but a pot used when at least one player goes all in and the other players continue betting; cannot be won by the player(s) who went all in
- **Suit**—a group of thirteen cards with values ranging from two to ace; the four types ranked lowest to highest are clubs, diamonds, hearts, and spades
- **Value**—refers to the second attribute in addition to suit that a card has; the fourteen values are 2-10, Jack, Queen, King, Ace, and Anteater
- **Game end**—game automatically ends when one player wins all the points or when all players agree to end the game via vote (computer players get no vote)

Betting Stages in a Hand (In Chronological Order)

1. **Preflop**—no cards on the board; betting starts with the small blind and the big blind, then proceeds to the player left of the big blind; each player can call, raise, or fold until all players have either met the highest bet made or folded
 - a. Note: During this stage, the only player who can check is the player who made the big blind, and this only occurs when everyone else has called or folded (i.e. not raised); the big blind player can then choose to either check (ending the preflop round) or raise (continuing the round, as all players then have to call, raise, or fold)
2. **Flop**—three cards on the board; each player can check (if no one else has bet), bet, call, raise, or fold until all players have either met the highest bet made or folded
3. **Turn** - After the flop, the dealer reveals the fourth community card; often where draws are complete and the pot size increases dramatically to force decisions
4. **River** - The last community cards is dealt in a game and where the final betting round begins
5. **Showdown** - the final stage of a hand in which all betting has concluded and only occurs when there are at least two players remaining after the final betting round or checking on the river. Typically, the last player to make a bet or raise on the final round is the first to reveal their hand, but if someone checks, the player to the left of the dealer reveals their hand first.

Note: If at any point all but one player has folded in the rounds leading up to the showdown, the one player left wins the pot.

Hand Rankings

1. **Royal flush**—the player has the 10, Jack, Queen, King, and Ace of all one suit
2. **Straight flush**—the player has five cards of consecutive value all of the same suit (i.e. a straight all in the same suit)
3. **Anteater Pair**—the player holds two anteaters in the hole cards
4. **Four of a kind**—the player has four cards of the same rank and one of another rank
5. **Full house** - the hand has three cards of one rank and two cards of another rank
6. **Flush** - a hand contains five cards of sequential order, not necessarily in consecutive order
7. **Straight** - a five-card hand containing sequential ranks, not considering the suits
8. **Three of a kind** - consisting of three cards of the same rank with two unmatched cards
9. **Two pair** - contains two cards of one rank, two cards another rank, and one card of a third rank
10. **Pair** - consists of two cards of the rank and three unrelated or unmatched cards
11. **High card** - contains none of the hands above, the hand is determined by the highest ranking card

The royal flush is the highest-ranking hand possible in the game

Moves

- **All-in**– Wager all of a player's remaining points on a single hand. The player can't take any more betting actions unless they win additional points in any side pots.
- **Bet**– The first voluntary wage made in a betting round.
- **Call**– Match the current highest bet made by the previous player. Only available when a preceding player has made a bet or raise during a betting round.
- **Check**– Pass the action to the next player without wagering. Only available if no player has bet before you in that round.
- **Fold**– Discarding your current hand and forfeit any points contributed in the pot
- **Raise**– Increase the current highest bet, requiring other players to match the new amount to keep their hand. Only available after a bet has been made.

1. Overview and Tutorial

1.1 Anteater Poker Rules/Gameplay

The user should first open the program by calling the executable file as specified in Section 2. Upon opening the program, the user will see a starting menu.

A player can choose to either be a host of a game or join an existing game within the starting menu. If they choose to be a host, they will set the number of players expected to join the game, the number of computer players/their positions, their own position, the number of points each player will start with, and the password for the game. If they choose to join as a non-hosting player, they will enter the password and the position they want to be at (one that is not already taken). The game will start once the number of players in the room equals the number specified by the host.

All players will start with the number of points specified by the host during game setup. In each round, they can gain or lose points by making bets. The game goes in order of the preflop, the flop, the turn, the river, and the showdown (see the glossary for more information). Betting in a round (except the showdown) continues until either players have all matched the highest bet or only one player is left. A round ends when either one player is left, who then takes all the points in the pot, or there are two or more players left at the showdown. In the event of a showdown, the player with the highest-ranking hand wins the pot. In the event of a tie, all tied players split the pot.

Players continue playing the game until the game is ended by the players (see "game end" in the glossary for more details) or until one player has a non-zero amount of points left at the end of a round.

1.2 Usage Scenarios

Start menu (host view, text version)

```
=====Welcome to Anteater Poker!=====
Are you the host (y/n)? y
How many players do you want?: 4
Of those players, how many do you want to be computer players?: 2
What position do want to be at (available positions: 1, 2, 3, 4)? 1
What position do you want computer 1 to be at (available positions: 2, 3, 4)? 3
What position do you want computer 2 to be at (available positions: 2, 4)? 4
Set the number of points each player will start with: 200
Okay! Starting game...
```

Start menu (non-host view, text version)

```
=====Welcome to Anteater Poker!=====
Are you the host (y/n)? n
What is the password for the game you want to join?: myPassword
Okay! Password recognized
What position do want to be at (available positions: 2)? 2
Okay! Starting game...|
```

Making a move (text version)

```
=====Round: Flop=====
=====Player 1's Cards=====
=====Hole Cards=====
Card 1: 7♦      Card 2: 5♣
=====Community Cards=====
Card 1: 9♦      Card 2: 7♣      Card 3: J♦      Card 4: ???      Card 5: ???
Current points: 100
Current amount wagered: 20
Current bet to call: 5 points
Enter bet, or type 'fold' to fold: 5
```

Help menu (text version)

```
=====Help Menu: How to Play=====
There can be up to five rounds in one hand: The preflop, the flop, the turn, the river, and the
showdown

Ranking of hands:
1. Royal flush
2. Straight flush
3. Anteater Pair
4. Four of a kind
5. Full house
6. Flush
7. Straight
8. Three pair
9. Two pair
10. Pair
11. High card

User actions: Player can type 'check' to check, a bet amount at or above the bet to call to
call/raise, or type 'fold' to fold
Anteater card: There are two anteater cards in the deck. If an anteater card is in the
community cards, it essentially does nothing. When a player holds two anteaters in the hole
cards, in which case it ranks between a royal and straight flush and four of a kind

=====End Help Menu=====
```

End of round

```
=====End of Round=====
Player 2 wins the main pot with: Full House!
Player 2 wins main pot value of: 40 points
Scores:
Player 1: 80 points      Player 2: 120 points      Player 3: 100 points      Player 4:
100 points
All players are still in the game.
Starting next round...
```

1.3 Goals

To create a fully functional poker game capable of supporting both human and computer players, recognizing valid moves, determining the winning five-card hand, and executing other functions and features necessary for the experience.

1.4 Features

The program will be able to support the following gamemodes with

- Player vs Computer (computer bots can be added to the game if desired)
- Player vs Player (a game with all human players)

Additional features:

- During the start menu, the host can set the amount of points each player starts with
- There are two anteater cards in the deck. If an anteater card is in the community cards, it essentially does nothing. When a player holds two anteaters in the hole cards, in which case it ranks between a royal and straight flush and four of a kind

2. Installation

2.1 System requirements

The user must satisfy the following conditions to correctly install the software:

- The user must have enough storage space on their computer to download the program
- The user must be able to access and use the command line to start the program
- The user must have RHEL8 64-bit Linux
- The user must have access to standard C, GTK 3.0, C++, and GTK++ library
- The user must be able to log onto a UCI EECS server for proper TCP/IP execution

2.2 Setup and configuration

Download the package. Using the command line, go into the installed folder.

For the binary package, run the executable program on the command line as follows (more details are in parentheses):

```
% gtar xvzf Poker_V1.0.tar.gz (extract contents into current directory)
% evince poker/doc/Poker_UserManual.pdf (for reading documentation, if desired)
% poker/bin/poker (run executable file)
```

For the source package, build the software using the following command line instructions (more details are in parentheses):

```
% gtar xvzf Poker_V1.0_src.tar.gz (extract contents into current directory)
% evince poker/doc/Poker_UserManual.pdf (for reading documentation, if desired)
```

```
% cd poker (go into chess directory)
% make (make executable file)
% make test (for testing code, if desired)
% make clean (to remove executables once finished, if desired)
```

2.3 Uninstalling

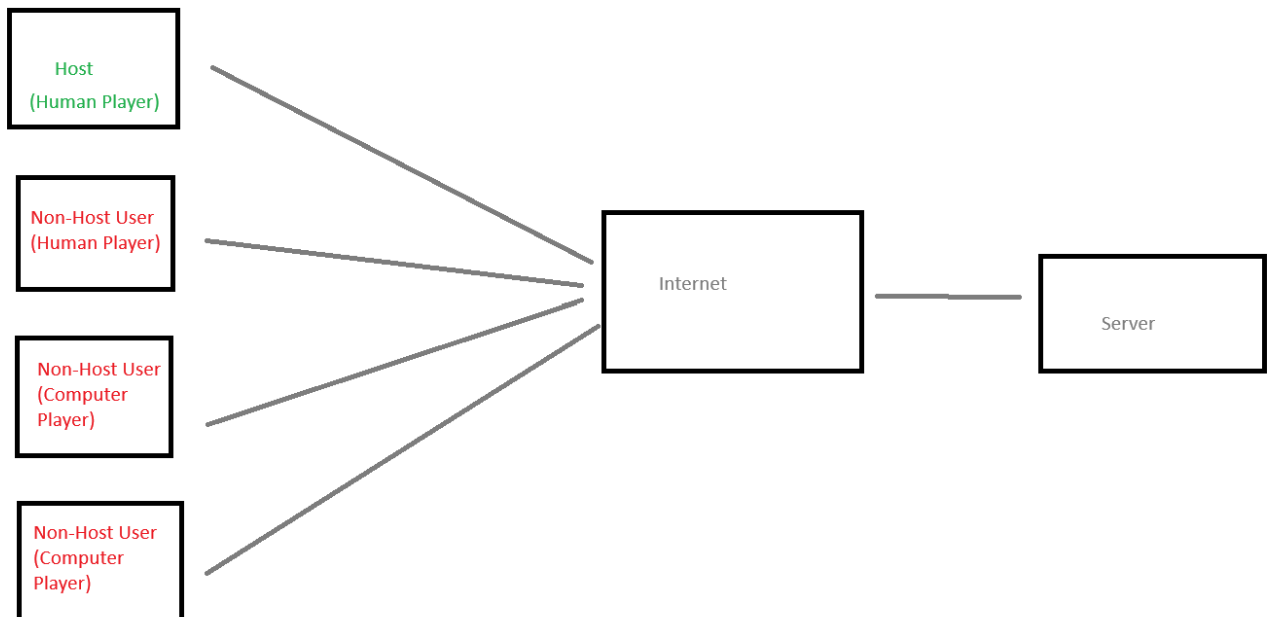
Through the command line, the user can uninstall the program by calling *make clean* (see above), then going one directory above the installed directory and calling *rmdir -rf PokerDirectoryName*, with *PokerDirectoryName* being the name of the directory that was downloaded by the user (the user may have decided to rename the directory, so be sure to check its current name).

3. Functions and Features

The full list of diagrams can be found under “Section 1.2 Usage Scenarios”.

3.1 Detailed description of client and server communication

Diagram of programs and network communication with Poker clients and Poker server (example case):



The poker server will be on a UCI EECS server, and it will use TCP/IP to handle communication between players and run the game.

3.2 Detailed description of dealer choice and card distribution

The dealer will be determined by the player positions chosen at the beginning of the game. It will start with player 1 and move to player 2, player 3, etc. until all players have had a chance to deal, at which point the deal will return to player 1.

User input:

The user will be prompted when it is their turn to deal. They will choose how many times to shuffle the cards by typing “shuffle” (text version) or pressing the “shuffle” button (GUI version) as many times as desired. Then, they will type or press “deal” to deal out cards according to Texas Hold’em rules.

Program output:

The program will indicate when it is a player’s turn to deal and when a deal has been completed.

Image (Player 1’s perspective, their deal, text interface version):

```
=====Start of Round=====
It is your turn to deal! Type 'shuffle' to shuffle the deck. Continue typing 'shuffle' as many times
as you would like, then type 'deal' to deal out the cards.
Would you like to shuffle the cards or deal the cards?: shuffle
Shuffling cards...
Would you like to shuffle the cards or deal the cards?: shuffle
Shuffling cards...
Would you like to shuffle the cards or deal the cards?: deal
Dealing cards...
Player 2 and Player 3 have made their blind and big blind wagers...
=====Round: Preflop=====
Big Blind (Submitting x to the pot): Player 2
Little Blind (Submitting 0.5x to the pot): Player 3
=====Player 1's Cards=====
=====Hole Cards=====
Card 1: 7 ♦      Card 2: 5 ♣
=====Community Cards=====
Card 1: ???      Card 2: ???      Card 3: ???      Card 4: ???      Card 5: ???
```

3.3 Detailed description of poker game integration

The game will integrate a user interface (text/graphics-based), TCP/IP communication protocols between the server and client, and the code used to run the game. The server will handle the logistics of the game, while the client-side computers will handle displaying the information from the server to the user.

User input

- Selects either check/bet or call/raise.
- If raise or bet is selected
 - Enter a number of points

Program output

- Assigns little blind and big blind before the hand, extracting corresponding amounts from their accounts
- Prompt Menu including {(Check, Bet, Fold, All-in), (Call, Raise, Fold, All-in)}
 - If no active betting amount exists (betting amount is equal to the default), prompt check
 - Extract default amount from the user's account
 - If active bet exists, prompt call
 - If the user's balance is lesser than the active bet, calling forces all their balance into the pot
 - If no raise has been made and it's the little blind's turn, extract the sufficient amount to match the bet.
 - Bet
 - Bet: If first user betting, require an opening bet greater than or equal to 10 points
 - Reprompt if false
 - Extract the corresponding amount from the user's account
 - Raise: Require a bet greater than the active bet
 - Reprompt if false
 - Extract the corresponding amount from the user's account
 - All-in
 - Automatically applied if the active bet trumps the user's balance.
 - Otherwise, extracts user's entire balance and submits it to the pot
 - Fold
 - The points remain in the pot, cards are unrevealed, and the player is inactive for the remainder of the hand.

Image:

```
=====Round: Flop=====
=====Player 1's Cards=====
=====Hole Cards=====
Card 1: 7 ♦      Card 2: 5 ♣
=====Community Cards=====
Card 1: 9 ♦      Card 2: 7 ♣      Card 3: J ♦      Card 4: ???      Card 5: ???
Current points: 100
Current amount wagered: 20
Current bet to call: 5 points
```

Options: Call (5), Raise, All-In, or Fold

// entered raise

Enter your action: Raise

// reprompting

Enter amount to raise (must be > 5): 3

Error: Raise must be a greater amount than the current bet. Try again.

Enter amount to raise (must be > 5): 15

Confirmed. 15 points extracted from account.

// entered call

Enter your action: Call

Confirmed. 5 points extracted from account to match the point bet.

// entered fold

Enter your action: Fold

Points update: 100 points remain (20 points remain in the pot)

Exiting till the next round...

Troubleshooting: Error Messages

Differing Scenarios:

- "ERROR: Failed to connect to server"
- "ERROR: Failed to connect all players"
- "ERROR: Failed to recognize winning hand"
- "ERROR: Failed to deal out another card to player"
- "ERROR: Failed to fill extra seats with bots"
- "ERROR: Failed to update the points of the player"

(Error messages are subject to change)

Contact Information

If you have any questions, comments, or concerns about this software, please contact the creators using the following email addresses:

- sbebal@uci.edu
- amchinn@uci.edu
- alonsojd@uci.edu
- edistanc@uci.edu
- lehowe@uci.edu
- samuek16@uci.edu
- davijl18@uci.edu

Note: Any email pertaining to this software sent to one or more of the addresses above may be forwarded to any or all of the other emails above.

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